Grade 9 Mathematics — Ganita Manjari

Chapter 12: Quadrilaterals | Exercise Set 12.1 Solutions

Question 1

Let $ABCD$ be a quadrilateral.
(i) List all sides of $ABCD$ adjacent to side AB. List all sides opposite to AB.
(ii) List all angles of $ABCD$ adjacent to $\angle A$. List all angles opposite to $\angle A$.
(iii) Define a pair of opposite sides and a pair of opposite angles without using the names of the vertices.

SOLUTION 1

- **(i) Sides adjacent and opposite to side AB:**
  - **Adjacent Sides:** BC and AD *(Both share endpoints B and A with side AB).*
  - **Opposite Side:** CD *(Lies directly across AB and shares no common vertex).*
- **(ii) Angles adjacent and opposite to $\angle A$:**
  - **Adjacent Angles:** $\angle B$ and $\angle D$ *(Angles that lie on the arms AB and AD meeting at vertex A).*
  - **Opposite Angle:** $\angle C$ *(Sits diagonally across from $\angle A$ with no common arm).*
- **(iii) Definitions without using vertex names:**
  - **Pair of Opposite Sides:** Two line segments forming the boundary of a quadrilateral that **do not share any common vertex (endpoint)**.
  - **Pair of Opposite Angles:** Two interior angles of a quadrilateral whose vertices **do not share a common side (arm)**.

Question 2

You have used internal angles of quadrilaterals, but they too require an exact definition... Precisely define the internal angle of a quadrilateral at a given vertex. Your answer should work for a non-convex quadrilateral too. (Hint: use the opposite vertex as well.)

SOLUTION 2

> 💡 **The Masterstroke Definition**
>
> At any given vertex of a quadrilateral, the **internal angle** is defined as the angle subtended between the two adjacent sides meeting at that vertex, measured through the region containing the interior of the polygon and oriented toward the opposite vertices.

Key Concept Breakdown:

- For any vertex V, the two meeting sides form two angles (one interior and one exterior, summing to 360°).
- The **internal angle** is always the angle facing the interior region where the structural diagonals and opposite vertices lie.
- *Universal Applicability:* This definition holds true for both **convex** (interior angle < 180°) and **non-convex** / **concave** quadrilaterals (where an interior reflex angle > 180° points inward).

Question 3

In a quadrilateral $ABCD$, suppose $AB \parallel DC$. Can $ABCD$ be non-convex? What if we instead assume $AB = CD$? What if we instead assume $\angle A = \angle C$?

SOLUTION 3

Case 1	$AB \parallel DC$	**Yes, it can be non-convex!** *Reason:* An arrowhead (dart) shape can have top and bottom edges parallel ($AB \parallel DC$) while one vertex is indented inward toward the interior.
Case 2	$AB = CD$	**Yes, it can be non-convex!** *Reason:* Having two opposite sides equal in length does not force the quadrilateral to be flat/convex; one vertex can be pushed inwards maintaining side lengths.
Case 3	$\angle A = \angle C$	**Yes, it can be non-convex!** *Reason:* A concave quadrilateral (such as a kite folded inward) can easily have one pair of opposite angles equal while remaining non-convex.

Question 4

Consider three non-collinear points A, B, C and draw the lines AB, BC, CA. For every possible location of point D in the plane outside these lines, decide if $ABCD$ is self-intersecting, non-convex, or convex. (Hint: the three lines divide the plane into 7 regions.)

SOLUTION 4 — THE 7-REGION MAGIC MAP

Extending the three lines AB, BC, CA partitions the infinite 2D plane into **7 regions** (1 central triangular region + 6 external regions):

- **Convex Regions (3 Vertex-Opposite Wedges):** When D is placed in the regions lying vertically opposite to the vertices of $\triangle ABC$, the quadrilateral $ABCD$ forms a clean **Convex** shape.
- **Non-Convex / Concave Regions (3 Side Regions):** When D is placed adjacent to the sides of $\triangle ABC$, one of the vertices of $\triangle ABC$ indents inward, forming a **Non-Convex** (arrowhead) quadrilateral.
- **Self-Intersecting Regions (Cross-over Zones):** Placing D in specific opposite cross-sectors forces the boundary lines to cross each other, forming a **Self-Intersecting** figure-eight quadrilateral.

Question 5

Can a quadrilateral be both self-intersecting and non-planar?

SOLUTION 5

Answer: YES, absolutely!

Explanation:

- **Planar:** Lies completely on a single 2D plane.
- **Self-Intersecting:** Two of its non-adjacent edges cross over each other.
- If you take a 3D wire model of a self-intersecting quadrilateral and lift one vertex out of the plane (skewing it in 3D space), the shape becomes **both non-planar and self-intersecting simultaneously**.

🌟 Interactive "Movable" Classroom Activity (Making Math Come Alive!)

Take 4 ice-cream sticks (or plastic strips) connected loosely at the vertices with push-pins or brass fasteners.
• **Move the joints:** Watch how a simple rectangle shifts into a parallelogram.
• **Push one vertex inward:** Instantly demonstrate how a convex shape becomes non-convex (concave) while keeping side lengths equal!